G10S2 AP Calculus Review Practice Answers

PA Resources, Hans 10-1

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Calc MCQ

1. A
2. A
3. A
4. B
5. A
6. A
7. E
8. A
9. B
10. A
11. A
12. A
13. B
14. A
15. A

Calc FRQ

Set 1: No Calculator

1. (a) $f'(x) = e^{-x}(1-x) = 0 \implies x = 1$, and $f''(1) = -e^{-1} < 0$, so a local maximum at $x = 1$.

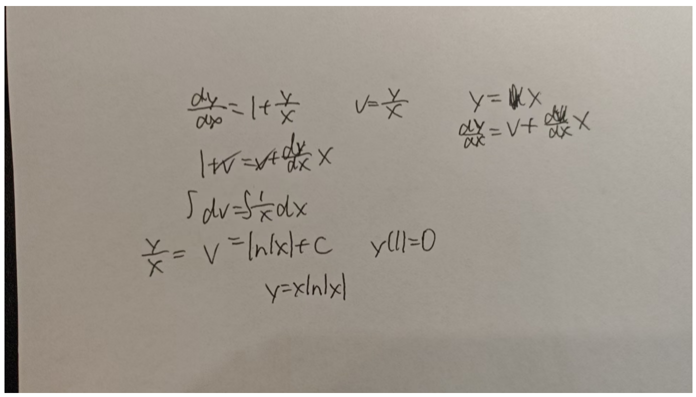
   (b) $\lim_{x\to\infty} x\, f(x) = \lim_{x\to\infty} x^2 e^{-x} = 0$.

2. By parts: $\int_0^{\pi} x\cos x\,dx = [x\sin x]_0^{\pi} - \int_0^{\pi} \sin x\,dx = 0 - 2 = -2.$

3. Ratio test:

$$\lim_{n\to\infty} \frac{(n+1)/2^{n+1}}{n/2^n} = \tfrac{1}{2} \implies R = 2, \quad I = (-1, 3).$$

4. Patrick:



Set 2: No Calculator

1. Solve $\sqrt{x} = x/2 \implies x = 0, 4$. Then

$$A = \int_0^4 \left(\sqrt{x} - \tfrac{x}{2}\right) dx = \left[\tfrac{2}{3}x^{3/2} - \tfrac{1}{4}x^2\right]_0^4 = \tfrac{16}{3} - 4 = \tfrac{4}{3}.$$

2. Shells about y-axis: $x \in [0, 2]$, $h = 4 - x^2$,

$$V = 2\pi \int_0^2 x(4 - x^2)\,dx = 2\pi\left[2x^2 - \tfrac{x^4}{4}\right]_0^2 = 2\pi(8 - 4) = 8\pi.$$

3. (a) $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \left.\dfrac{e^t(\sin t + \cos t)}{e^t(\cos t - \sin t)}\right|_{t=0} = 1.$

   (b)

$$\int_0^{\ln 2} y\,\frac{dx}{dt}\,dt = \int_0^{\ln 2} e^{2t}(\sin t\cos t - \sin^2 t)\,dt = \sin(2\ln 2) - \tfrac{3}{4}.$$

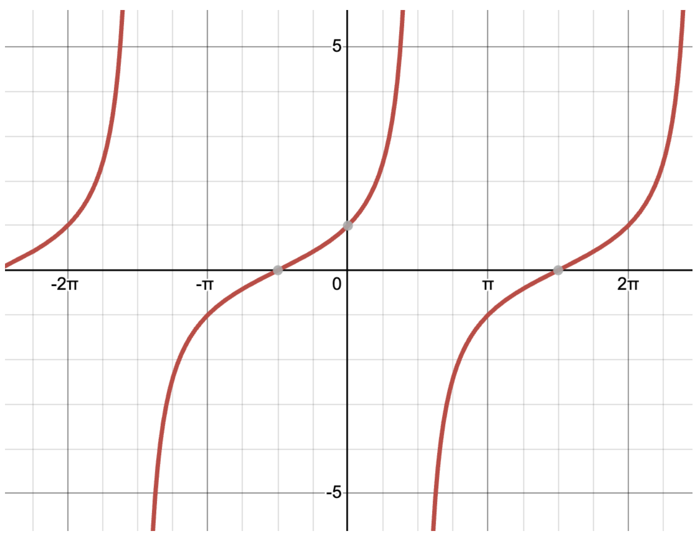
4. Alternating $1/n^2$ series converges absolutely (p-series, $p = 2 > 1$).

Set 3: Calculator Allowed

1. (a) Newton: $x_1 = x_0 - \frac{\cos x_0 - x_0}{-\sin x_0 - 1} = 0.5 + \frac{0.37758}{1.47943} \approx 0.7552$.
   (b) Error: $|\cos(0.7552) - 0.7552| \approx 0.0262$.
2. $\displaystyle\int_1^3 e^{-x^2}\,dx \approx 0.1394\,.$
3. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots \implies \ln 1.5 \approx 0.5 - 0.125 + 0.02083 = 0.3958.$
4. Solve $\ln x = x - 2$ numerically: $x \approx 3.15$.

Precalc

1. Label all intercepts, asymptotes, and indicate periodicity.



2. $\log_3 \log_2 m = \log_3(3\log_8 m) = 1 + 2\log_9(\log_8 m) = 1 + 2 \cdot 2024 = 4049.$
3. **Solve on** $[0, 2\pi]$:
   1. $\sin(3x) - \sin x = \sqrt{3}\cos(2x)$.

$$2\cos 2x \sin x = \sqrt{3}\cos 2x \implies \begin{cases} \cos 2x = 0 \implies x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \\ 2\sin x - \sqrt{3} = 0 \implies x = \frac{\pi}{3}, \frac{2\pi}{3}. \end{cases}$$

   2. $1 - \cos x = 2\sin^2 x$.

$$2\cos^2 x - \cos x - 1 = 0 \implies \cos x = 1 \text{ or } -\tfrac{1}{2} \implies x = 0, 2\pi,\ \tfrac{2\pi}{3},\ \tfrac{4\pi}{3}.$$

4.

$$P(t) = 60 + 20\sin\Big(\frac{\pi}{12}t + \frac{\pi}{6}\Big), \qquad T = \frac{2\pi}{\omega} = 24 \text{ days.}$$

Minima occur at $t = 16 + 24k$, maxima at $t = 28 + 24k$, so in 365 d you get

$$\Big\lfloor \frac{365 - 16}{24} \Big\rfloor + 1 = 15$$

full buy-low–sell-high cycles. Each cycle doubles your capital:

$$C_{365} = 700 \cdot 2^{15} = 22\,937\,600, \qquad \text{net profit} = 22\,937\,600 - 700 = 22\,936\,900.$$

5. **Simplify (for $x \in [2\pi, 3\pi)$):**

   1. $\arccos(\sin x) = \left|\frac{5\pi}{2} - x\right|$.
   2. $\tan\big(\arccos x\big) = \dfrac{\sqrt{1-x^2}}{x}, \quad x \in (-1,1) \setminus \{0\}$.
   3. $\cos\big(\arcsin x\big) = \sqrt{1-x^2}, \quad x \in [-1,1]$.

6. The second half becomes

$$\sin(A + \pi/4) = \sin(B + \pi/4).$$

If $(A + \pi/4) + (B + \pi/4) = \pi$, then $C = \pi/2$, contradiction. Hence, $A = B \in (0, \pi/2)$ and $C = \pi - 2A$.

Then, $\cos C = \frac{\sqrt{2}}{2} \sin(A + \pi/4) > 0$ and so $C < \pi/2$. Hence, $A = B \in (\pi/4, \pi/2)$, giving

$$\begin{aligned}\cos C &= -\cos(2A) = -\sin(2A + \pi/2) = -2\sin(A + \pi/4)\cos(A + \pi/4)\\ &= -2\sqrt{2}\cos C \cdot (-\sqrt{1 - (\sqrt{2}\cos C)^2}).\end{aligned}$$

Solving, $\cos C = \frac{\sqrt{7}}{4}$.

7.

10.（本题满分 20 分）在平面直角坐标系中，双曲线$\Gamma: x^2-y^2=1$的右顶点为A．将圆心在y轴上，且与Γ的两支各恰有一个公共点的圆称为“好圆”．若两个好圆外切于点P，圆心距为d，求$\frac{d}{|PA|}$的所有可能的值．

解：考虑以$(0, y_0)$为圆心的好圆$\Omega_0: x^2+(y-y_0)^2=r_0^2\ (r_0>0)$．

由Ω_0与Γ的方程消去x，得关于y的二次方程

$$2y^2-2y_0y+y_0^2+1-r_0^2=0.$$

根据条件，该方程的判别式$\Delta=4y_0^2-8(y_0^2+1-r_0^2)=0$，因此$y_0^2=2r_0^2-2$．

…………5 分

对于外切于点P的两个好圆Ω_1, Ω_2，显然P在y轴上．设$P(0,h)$，Ω_1, Ω_2的半径分别为r_1, r_2，不妨设Ω_1, Ω_2的圆心分别为$(0, h+r_1)$, $(0, h-r_2)$，则有

$$(h+r_1)^2=2r_1^2-2,$$
$$(h-r_2)^2=2r_2^2-2.$$

两式相减得$2h(r_1+r_2)=r_1^2-r_2^2$，而$r_1+r_2>0$，故化简得$h=\frac{r_1-r_2}{2}$．

…………10 分

进而$\left(\frac{r_1-r_2}{2}+r_1\right)^2=2r_1^2-2$，整理得

$$r_1^2-6r_1r_2+r_2^2+8=0. \qquad ①$$

由于$d=r_1+r_2$，$A(1,0)$，$|PA|^2=h^2+1=\frac{(r_1-r_2)^2}{4}+1$，而①可等价地写为$2(r_1-r_2)^2+8=(r_1+r_2)^2$，即$8|PA|^2=d^2$，所以$\frac{d}{|PA|}=2\sqrt{2}$．　…………20 分